Article Title

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Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

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1 Introduction

The dentate gyrus (DG) is widely held to perform pattern separation, the transformation of similar inputs into more distinct neural representations, so that overlapping experiences can be encoded without interfering with one another (Yassa and Stark 2011; Rolls 2013; Hainmueller and Bartos 2020). This function is commonly attributed to the sparse, competitive activity of its principal excitatory cells, the granule cells (GCs): of the large GC population in the granular layer of the DG, one of the densest regions of the brain in cell bodies (Amaral et al. 2007), only a small fraction of cells fires in response to any given input arriving from the entorhinal cortex (EC), the main afferent to the DG (Kesner and Hopkins 2006; Chavlis et al. 2017).

The DG is also one of the few regions of the adult mammalian brain that sustains adult neurogenesis, continually generating new GCs throughout life (Altman and Das 1965; Ming and Song 2011). These cells mature over several weeks, passing through a critical window at roughly 4–7 weeks of age during which the immature granule cells (iGCs) are still only partially integrated into the circuit yet markedly more excitable than their mature counterparts (mGCs) (Espósito et al. 2005; Aimone et al. 2014; Denoth-Lippuner and Jessberger 2021). Enhancing neurogenesis has been shown to improve behavioral pattern separation (Sahay et al. 2011), showing that these cells are central to the DG's discriminative function.

How iGCs exert this influence at the level of the circuit, however, remains unsettled (Berdugo-Vega et al. 2023). One view treats them as encoders that integrate directly into memory representations alongside mGCs; another, gaining more recent support, views them instead as transient modulators that shape the activity of the surrounding network rather than encoding information themselves (Aimone 2016; Fölsz et al. 2023). Consistent with the latter, McHugh et al. (2022) showed *in vivo* that briefly activating immature adult-born GCs recruits inhibitory interneurons throughout the hippocampus and increases population sparsity in both the DG and CA3, pointing to a circuit-level mechanism by which a small population of new cells could regulate activity well beyond the DG. Similarly, Ko et al. (2025) showed that adult neurogenesis drives the reorganization of hippocampal engram circuitry as memories are consolidated, remodeling the DG–CA3–CA1 network so that its representations generalize over time. These results show that iGCs have important effects across the hippocampus network, not being restricted to a local function in the DG only.

Given the difficulty of experimentation in the DG and the complexity of information processed in the hippocampus, computational models have been invaluable for understanding the role of neurogenesis (Aimone 2016). Existing models of neurogenesis and pattern separation, however, are confined almost entirely to the DG (Chavlis et al. 2017; Kim and Lim 2024b; Yang et al. 2025), and so cannot investigate the influence of iGCs into CA3 and CA1.

This work aims to investigate whether the circuit-wide sparsification observed *in vivo* emerges from the known DG–CA3 microcircuitry has likewise not been examined in a biophysically detailed model.

2 Methods

2.1 Network architecture

The DG-CA3 network architecture, shown in Figure 1, was built primarily upon the models of Chavlis et al. (2017); Kim and Lim (2024a); Kopsick et al. (2024); Yang et al. (2025) and reproduced at a scale of $\frac{1}{500}$ of the rat hippocampus, the per-population cell counts being listed in Table 1. The neuron and synapse models, together with the cell counts and connectivity, were adapted from Hippocampome.org, an open-access knowledge base that consolidates multiple experimental sources on the hippocampal formation (Wheeler et al. 2023).

The model included the core excitatory and inhibitory cell types of the DG-CA3 network underlying pattern separation and completion. Although both regions contain many additional interneuron subtypes, several of which are also characterized in Hippocampome.org (Wheeler et al. 2023), this reduced set has been shown in previous computational models to be sufficient to support pattern separation (Chavlis et al. 2017; Kim and Lim 2024b; Yang et al. 2025). Synaptic connectivity between cell types and their connection probabilities were likewise derived from multiple sources and constrained to biologically plausible ranges. After the parameters had been gathered from Hippocampome.org, and the network had been assembled, a subset of neuron and synapse parameters, together with the selected connection probabilities, was adjusted within biologically reasonable bounds to reproduce the *in vivo* firing rate and population sparsity associated with each cell type (Online Resource 1, Table S1).

The network receives input from a population of ($N_{EC} = 400$) entorhinal cortex (EC) neurons (Amaral et al. 1990; Kim and Lim 2024a). In each simulation, the EC encodes a specific input pattern, represented by an active subpopulation comprising 10% of the EC neurons (McNaughton et al. 1991). Neurons not included in the active pattern remain silent throughout the simulation, whereas active neurons generate spikes according to a Poisson process with a mean firing rate of 40 Hz. EC neurons project their axons via the perforant path to neurons with dendrites in the molecular layer of the DG: including granule cells (GCs), mossy cells (MCs) (Scharfman and Myers 2013) and basket cells (BCs), as well as to neurons in CA3.

The GC population is split into mGCs and iGCs, the latter standing for the cells that adult neurogenesis produces during their characteristic 4–7 weeks electrophysiological window (Aimone et al. 2014). Of the $N_{GC} = 2000$ GCs (West et al. 1991), 5% are immature (Cameron and Mckay 2001), yielding $N_{mGC} = 1900$ and $N_{iGC} = 100$. To reflect the lamellar organization of the DG (Sloviter and Lømo 2012), the GCs are distributed across 20 lamellae, each containing 100 cells. Each GC connects to the BCs and CA3 neurons within its own lamella and also forms random non-lamellar excitatory connections with hilar perforant path-associated cells (HIPPs) and MCs.

iGCs are characterized by electrophysiological properties distinct from the mGCs that make them more excitable than the mGCs. Furthermore, to simulate their still-incomplete circuit integration, iGCs receive EC afferent inputs at only a percentage $x_{EC \to iGC}$ of the connection probability between the EC and mGCs, and also receive reduced inhibitory inputs from BCs and HIPPs (Table 3). Additionally, some iGCs are known to also directly inhibit mGCs (Luna et al. 2019), so the model includes an inhibitory synapse from iGCs onto mGCs. Because Hippocampome.org (Wheeler et al. 2023) does not characterize iGCs, six of the nine iGC parameters were fitted to electrophysiological data reported by Espósito et al. (2005). Parameter fitting was performed using the Nelder and Mead (1965) simplex method by minimizing the squared error between the model output and the number and timing of the spikes evoked by a range of depolarizing current steps. The membrane capacitance, resting potential and spike peak voltage were held fixed (Table 2).

BCs ($N_{BC} = 60$) are inhibitory cells that ensure sparse GC activation through winner-take-all competition among them (Coultrip et al. 1992; Chavlis et al. 2017; Kim and Lim 2024a), such that only the most active GCs remain active in a lamella, whereas

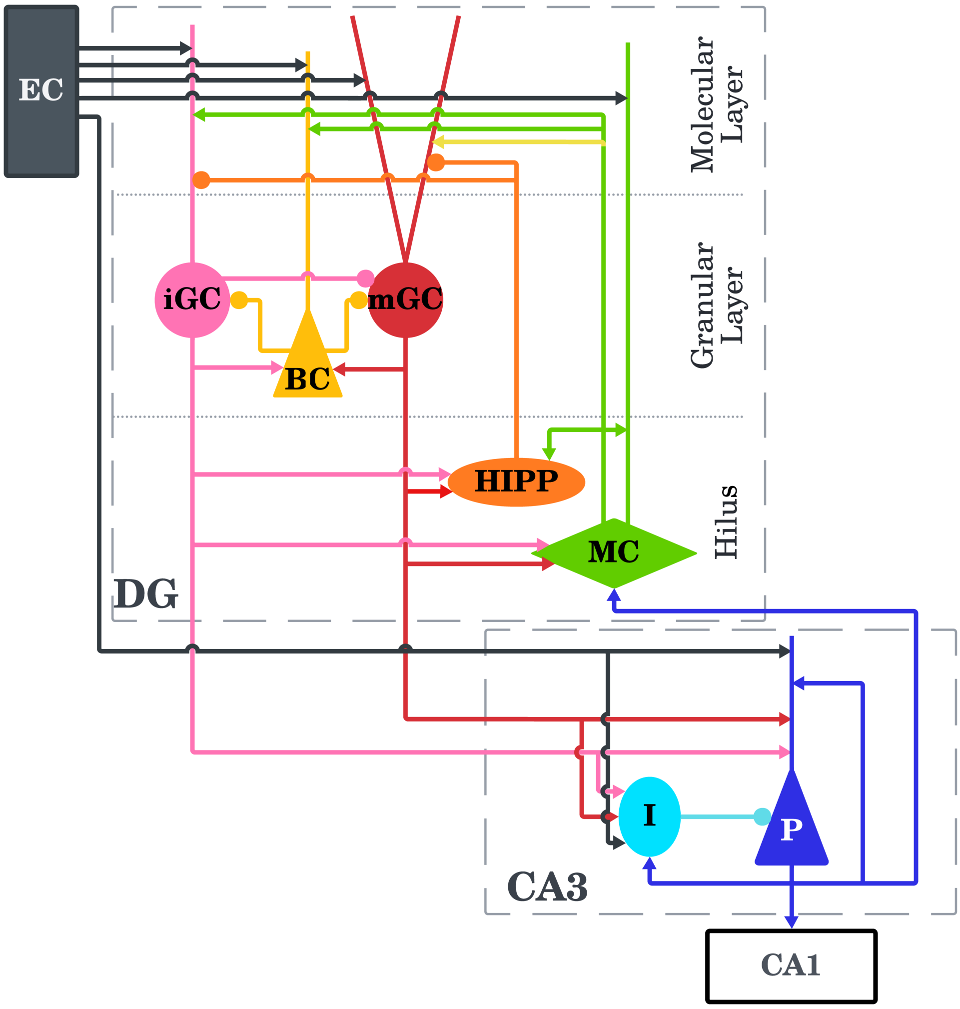

Fig. 1: DG–CA3 network architecture. Network input is provided by entorhinal cortex (EC) neurons. In the dentate gyrus (DG), the perforant path terminates in the molecular layer, where the dendrites of DG neurons are located. The granule cell layer comprises immature granule cells (iGCs), mature granule cells (mGCs), and basket cells (BCs), whereas the hilus contains mossy cells (MCs) and hilar perforant path-associated cells (HIPPs). The CA3 region comprises excitatory pyramidal cells (pCA3) and inhibitory interneurons (iCA3). Excitatory synapses are represented by arrows, whereas inhibitory synapses are represented by circles.

the others are suppressed. HIPPs ($N_{HIPP} = 60$) also contribute to the sparsity of GC activation by providing a more global inhibition to all GCs in a lamella. MCs ($N_{MC} = 100$) are excitatory hilar cells that receive excitation from GCs and project back to GCs, BCs, HIPPs and other MCs via cross-lamellar connections. Although direct MC to GC projections are excitatory, their net effect is generally inhibitory through the control of BCs and HIPPs (Myers and Scharfman 2009; Scharfman and Myers 2013).

The HIPP parameters provided by Hippocampome.org (Wheeler et al. 2023) reproduce a rebound burst firing, in which a burst of spikes occurs following the termination of the hyperpolarization. However, strong hyperpolarization produced excessively prolonged bursts that destabilized the network simulations. Therefore, the HIPP parameters were instead adopted from Modak and Chakravarthy (2018).

CA3, unlike the DG, does not follow a lamellar structure in the model, as it forms a far more integrative neural network (Pak et al. 2022; Watson et al. 2025). It holds $N_{pCA3} = 600$ pCA3 cells and $N_{iCA3} = 60$ iCA3 cells, the latter modeled after CA3 basket cells (Wheeler et al. 2023) so as to simplify the immense variety of inhibitory neurons present in CA3 (Kopsick et al. 2024). Both neuronal populations receive afferent inputs from the EC and from GCs, with iCA3 cells inhibiting pCA3 cells, which in turn excite the iCA3 cells and, crucially, wire recurrently among themselves, the substrate of the autoassociation and pattern-completion functions of CA3 (Rolls 2013; Kopsick et al. 2024). A portion of the pCA3 cells further feed back to the DG by contacting MCs, a projection that contributes to pattern separation (Myers and Scharfman 2011).

All simulations were run in Brian2 (Stimberg et al. 2019) with the fourth-order Runge-Kutta method at a fixed time step of 0.1 ms (Butcher 1996); the neuron and synapse parameters appear in Tables 2 and 3. For each input pattern the network was left 300 ms to reach a steady state, and its activity was then recorded over the following 1000 ms. Synaptic connections remained fixed throughout.

Cell	**EC**	**mGC**	**iGC**	**MC**	**HIPP**	**BC**	**pCA3**	**iCA3**
N	400	1900	100	100	60	60	600	60

Table 1: Neuronal population size (N).

2.2 Neuron model

Each neuron was reduced to a single compartment, with no explicit dendrites or axons, yet kept biologically plausible, and obeys the nine-parameter Izhikevich model (Izhikevich 2006, chapter 8). This model was chosen in this work and Hippocampome.org for being capable of reproducing the firing dynamics of neurons across a wide variety of conditions with a biological plausibility approaching that of the Hodgkin-Huxley model (Hodgkin and Huxley 1952), but at a far lower computational cost. Its dynamics are described by:

$$C_m \frac{dV_m}{dt} = k(V_m - V_r)(V_m - V_t) - u + I \tag{1}$$

$$\frac{du}{dt} = a[b(V_m - V_r) - u] \tag{2}$$

where V_m is the membrane potential, u the recovery variable, C_m the membrane capacitance, V_r the resting potential, V_t the threshold potential, I the total current into the neuron, and k, a and b constants that shape the neuron's dynamics. These continuous equations are complemented by a discrete rule for action-potential generation (Equation 3):

$$\text{if } V_m \geq V_{\text{peak}}, \quad \begin{cases} V_m \leftarrow V_{\text{min}} \\ u \leftarrow u + d \end{cases} \tag{3}$$

When V_m reaches the peak value V_{peak}, a spike is emitted, V_m is reset to the after-spike potential V_{min}, and the recovery variable is incremented by d, which makes the following spikes harder to elicit.

Cell	k (nS/mV)	a (ms^{-1})	b (nS)	d (pA)	C_m (pF)	V_r (mV)	V_t (mV)	V_{min} (mV)	V_{peak} (mV)
mGC	0.45	0.003	24.48	50	38	-77.4	-44.9	-66.47	15.49
iGC	0.139	0.002	-1.877	12.149	24.6	-63.66	-38.41	-48.2	83.5
MC	1.5	0.004	-20.84	117	258	-63.67	-37.11	-47.98	28.29
HIPP	0.01	0.004	-2	40.52	58.7	-70	-50	-75	90
BC	0.81	0.097	1.89	553	208	-61.02	-37.84	-36.23	14.08
pCA3	0.79	0.008	-10	588	186	-63.2	-33.6	-38.87	35.86
iCA3	1	0.004	-18.26	12	45	-57.51	-23.38	-47.56	18.45

Table 2: Izhikevich model parameters per cell type.

2.3 Synapse model

The synapse model follows the formulation of Senn et al. (2001) and Mongillo et al. (2008). It captures short-term depression, from neurotransmitter depletion, and facilitation, from calcium accumulation, acting over hundreds of milliseconds. Five parameters describe each synapse (Table 3): the maximum conductance in the absence of resource depletion g, the fraction of resources spent per spike U_{se}, the synaptic-current decay constant τ_d, the facilitation constant τ_f and the resource-recovery constant τ_r (Moradi et al. 2022). The same table also reports the connection probability P; although Hippocampome.org supplies these probabilities *in vivo*, the reduced scale of the network made it necessary to raise them so that activity could be sustained without losing sparseness.

Three state variables define the synapse: the utilization of synaptic resources (U), their recovery (R) and the fraction held in the active state (A). All resources start available, so that $U_{t_0} = 0$, $R_{t_0} = 1$ and $A_{t_0} = 0$. They evolve according to:

$$\frac{dU}{dt} = \frac{-U}{\tau_f} + U_{se}(1 - U_-)\delta(\Delta t_i) \tag{4}$$

$$\frac{dR}{dt} = \frac{1 - R - A}{\tau_r} - U_+ R_- \delta(\Delta t_i) \tag{5}$$

$$\frac{dA}{dt} = \frac{-A}{\tau_d} + U_+ R_- \delta(\Delta t_i) \tag{6}$$

where δ is the Dirac delta, equal to 1 only when $\Delta t_i = t - t_i = 0$, that is, at the instant of a synaptic event t_i; U_+ denotes the value of U just after the event and R_- the value of R just before it. The resulting synaptic current is:

$$I_{syn} = s \cdot A \cdot g \cdot (V_m - E) \tag{7}$$

where V_m is the postsynaptic membrane potential and E the reversal potential, set to $E_{inh} = -86\,\text{mV}$ for inhibitory synapses and $E_{exc} = 0\,\text{mV}$ for excitatory ones. The constant $s = 10$, applied uniformly to every synapse, is a scaling factor required by the reduced size of the network: with so few synapses relative to the rat hippocampus, the network would otherwise fall silent. The synaptic currents I_{syn} reaching a neuron are summed into the total current I of its membrane equation (Equation 1).

2.4 Pattern separation

Pattern separation was quantified following Kim and Lim (2024b). To probe its dependence on input similarity and on adult neurogenesis, 11 variants of the network were compared: a control without neurogenesis ($N_{mGC} = 2000$) and 10 with neurogenesis ($N_{mGC} = 1900$, $N_{iGC} = 100$) at increasing levels of EC→iGC excitability, as illustrated in Figure 2. This level, ranging from 10% to 100% in increments of 10%, fixes the EC to iGC connection probability as a percentage relative to that of the mGCs: with 100% matching the mGC connection probability, and 10% meaning one tenth of that probability, and so represents the still-incomplete dendritic integration of the immature cells. Because iGCs acquire their entorhinal input only gradually over the 4–7 week maturation window (Espósito et al. 2005; Aimone et al. 2014), no single value of this excitability is representative of the immature population; the full range was therefore swept, rather than assuming one level, with lower values corresponding to earlier, less integrated cells and higher values to more mature ones approaching the connectivity of the mGCs. Each variant was presented with 30 sets of 10 input patterns, a set being one randomly generated reference pattern together with 9 derivatives of it graded from 90% down to 10% similarity. An input pattern is a binary vector of length $N_{EC} = 400$ encoding the state of each EC neuron, its active neurons firing as Poisson processes at 40 Hz and exactly 10% (40 neurons) being active in every pattern. A pattern of similarity $X\%$ was obtained by keeping a random fraction of $X\%$ of the reference pattern's active neurons and moving the remaining $(100 - X)\%$ to previously inactive neurons, which preserves the 10% activity level.

Presynaptic	Postsynaptic	Connection	P (%)	g (nS)	τ_d (ms)	τ_r (ms)	τ_f (ms)	U
EC	mGC	Random	8	1.655	5.333	266.239	18.714	0.27
EC	iGC	Random	0.8–8*	1.655	5.333	266.239	18.714	0.27
EC	MC	Random	20	1.522	4.671	319.835	57.766	0.204
EC	BC	Random	25	1.406	3.849	144.415	48.2	0.214
EC	pCA3	Random	12	1.5	6.55	258.318	53.478	0.184
EC	iCA3	Random	12	2	3.602	257.468	35.904	0.21
mGC	MC	Random	40	2.713	5.347	98.583	27.479	0.151
mGC	HIPP	Random	10	1.805	5.181	362.814	48.986	0.15
mGC	BC	Lamellar	100	4.258	5.566	31.265	4.278	0.497
mGC	pCA3	Lamellar	80	1.5	6.657	278.286	78.584	0.155
mGC	iCA3	Lamellar	50	3	3.915	218.934	43.274	0.176
iGC	mGC	Random	10	0.524	3.915	218.934	43.274	0.176
iGC	MC	Random	40	2.713	5.347	98.583	27.479	0.151
iGC	HIPP	Random	10	1.805	5.181	362.814	48.986	0.15
iGC	BC	Lamellar	100	4.258	5.566	31.265	4.278	0.497
iGC	pCA3	Lamellar	80	1.5	6.657	278.286	78.584	0.155
iGC	iCA3	Lamellar	50	3	3.915	218.934	43.274	0.176
MC	mGC	Cross-lamellar	0.2	1.894	5.357	166.162	20.224	0.304
MC	iGC	Cross-lamellar	0.2	1.894	5.357	166.162	20.224	0.304
MC	MC	Cross-lamellar	5	0.068	4.257	149.329	71.642	0.245
MC	HIPP	Cross-lamellar	50	2.376	4.824	358.431	54.872	0.181
MC	BC	Cross-lamellar	100	1.996	3.396	117.365	69.316	0.255
HIPP	mGC	Random	30	3.202	8.935	159.143	10.396	0.278
HIPP	iGC	Random	10	3.202	8.935	159.143	10.396	0.278
BC	mGC	Lamellar	100	4.351	3.543	103.876	12.347	0.332
BC	iGC	Lamellar	10	4.351	3.543	103.876	12.347	0.332
pCA3	pCA3	Random	2.5	0.9	9.516	278.258	27.513	0.172
pCA3	MC	Lamellar	10	2.035	4.297	359.116	40.457	0.236
pCA3	iCA3	Random	20	2	4.525	275.604	23.321	0.189
iCA3	pCA3	Random	30	2	7.793	416.282	20.63	0.203

Table 3: Parameters of the synapses between neuronal populations. Random connections occur between all cells of both populations; lamellar connections occur between cells of the same lamella; cross-lamellar connections occur between cells of one lamella and all others. The connection probability P refers to the percentage of connections between neuronal populations according to the connection condition. * iGCs were simulated with multiple different EC$\rightarrow$iGC excitability levels.

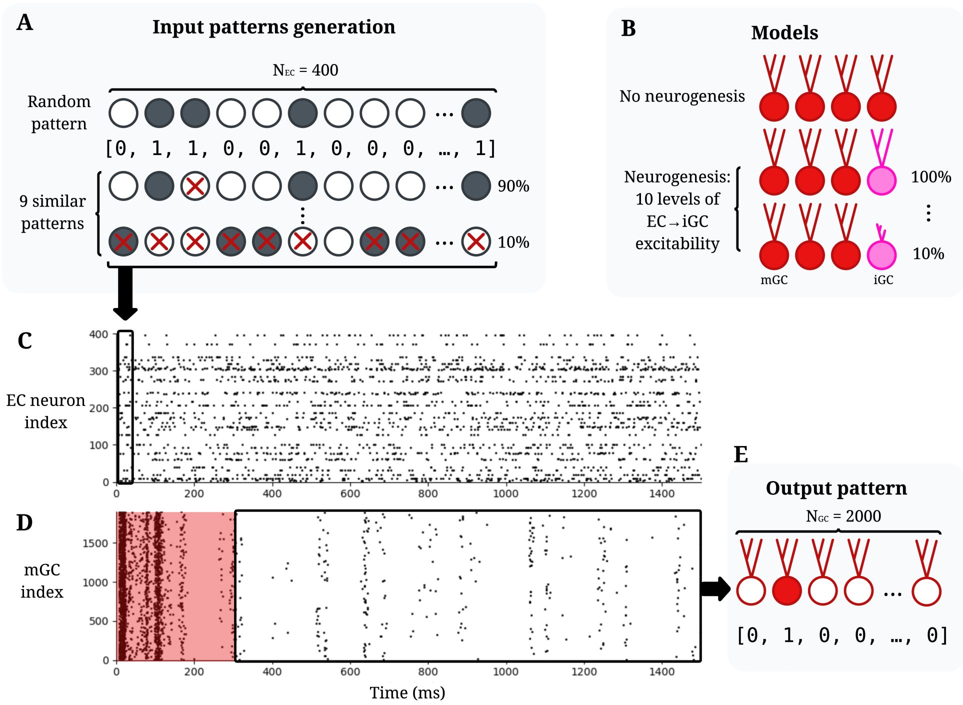

Fig. 2: Diagram of the input and output pattern generation and recall protocol. **A.** Input patterns are generated in sets of 10 patterns of size $N_{EC} = 400$; the first is generated randomly and the remaining ones are generated from it at different levels of similarity by randomly activating and deactivating neurons. **B.** 11 model variants are simulated: one without neurogenesis and 10 with neurogenesis at different excitability levels, from 100% down to 10% excitability. **C.** Spike trace of EC neurons during a simulation; each generated pattern corresponds to the activation of EC neurons, with active neurons spiking at a frequency of 40 Hz; each of the 11 models is exposed to 30 sets of 10 input patterns, totaling 300 simulations of 1300 ms each. **D.** Spike trace of mGCs during a simulation; the first 300 ms are discarded while the network stabilizes; the remainder corresponds to the output pattern. **E.** The output pattern, illustrated here for the mGC population, consists of a binary vector with value 1 for the index of each neuron that spiked at least once during the 1000 ms interval.

Each pattern is simulated for 1300 ms of simulation time, with the first 300 ms discarded for the network to stabilize. Separation was then read from the overlap between the activity patterns at the input (EC cells) and the output (GCs or pCA3 cells). The output pattern is a binary vector of length N, the number of cells in the principal cell population of an area, be it GCs or pCA3s, whose entries are 1 for neurons that fired at least once during the 1000 ms window and 0 otherwise[1]. For a pair of patterns $A^{(l)}$ and $B^{(l)}$, with $l \in \{in, out\}$ for input and output, the pattern distance $D_p^{(l)}$ is defined as:

$$D_p^{(l)} = \frac{O^{(l)}}{D_a^{(l)}} \tag{8}$$

where $O^{(l)}$ is the orthogonalization degree and $D_a^{(l)}$ the mean activation of the pair, itself the arithmetic mean of the activation degrees of $A^{(l)}$ and $B^{(l)}$:

$$D_a^{(l)} = \frac{D_a^{(A^{(l)})} + D_a^{(B^{(l)})}}{2} \tag{9}$$

the activation degree of a pattern being its fraction of active neurons. The orthogonalization $O^{(l)}$, which gauges the dissimilarity between the patterns, follows from the Pearson correlation coefficient $\rho^{(l)}$:

$$O^{(l)} = \frac{1 - \rho^{(l)}}{2} \tag{10}$$

Writing $\{a_i^{(l)}\}$ and $\{b_i^{(l)}\}$ $(i = 1, \ldots, N_l)$ for the binary states of the i-th cell in $A^{(l)}$ and $B^{(l)}$ $(l \in \{in, out\})$, this coefficient is:

$$\rho^{(l)} = \frac{\sum_{i=1}^{N_l} \Delta a_i^{(l)} \cdot \Delta b_i^{(l)}}{\sqrt{\sum_{i=1}^{N_l} \left(\Delta a_i^{(l)}\right)^2} \sqrt{\sum_{i=1}^{N_l} \left(\Delta b_i^{(l)}\right)^2}} \tag{11}$$

with $\Delta a_i^{(l)} = a_i^{(l)} - \langle a^{(l)} \rangle$, $\Delta b_i^{(l)} = b_i^{(l)} - \langle b^{(l)} \rangle$ and $\langle \ldots \rangle$ the mean over the whole population. Since $\rho^{(l)}$ ranges over $[-1, 1]$, the orthogonalization $O^{(l)}$ ranges over $[0, 1]$ (Equation 10).

From the input ($D_p^{(in)}$) and output ($D_p^{(out)}$) distances, the degree of pattern separation $\mathcal{S}_D$ is taken as their ratio:

$$\mathcal{S}_D = \frac{D_p^{(out)}}{D_p^{(in)}} \tag{12}$$

A value $\mathcal{S}_D > 1$ means that the output patterns are more distinct than the inputs, that is, pattern separation, whereas $\mathcal{S}_D < 1$ means the opposite, a convergence in which the outputs grow more alike.

[1] Computing pattern separation from graded spike counts instead of binary activation was also considered, but it yields nearly identical results (Pearson $r = 0.99$).

2.5 Population sparsity

Population sparsity measures how unevenly spiking is distributed across a population, being high when activity is concentrated in a few cells and low when it is spread evenly. Following McHugh et al. (2022), it was quantified with two complementary indices computed on the per-cell spike counts accumulated over the recording window. For a population of N cells with spike count vector x sorted in ascending order ($x_1 \leq \cdots \leq x_N$), the Gini index (Gini 1921) is:

$$S_{Gini} = \frac{\sum_{i=1}^{N}(2i - N - 1)\, x_i}{N \sum_{i=1}^{N} x_i} \tag{13}$$

where i is the rank of each cell in the sorted vector. As a cross-validation, the Hoyer measure (Hoyer 2004) was also computed:

$$S_{Hoyer} = \frac{\sqrt{N} - \left(\sum_{i=1}^{N} |x_i|\right) / \sqrt{\sum_{i=1}^{N} x_i^2}}{\sqrt{N} - 1} \tag{14}$$

Both indices range over $[0, 1]$, with larger values denoting sparser activity. They were computed for the mGC, iGC and pCA3 populations and for the combined granule-cell population (mGCs and iGCs together), and averaged over all patterns and trials, with the standard error of the mean reported as the error.

2.6 iGC activation and downstream circuit effects

McHugh et al. (2022) showed *in vivo* that briefly activating immature (4–7 weeks) adult-born GCs recruits hippocampal interneurons and increases population sparsity in the DG and CA3. To test whether the model reproduces these circuit effects, iGCs were activated directly through current injection and the response of every population was measured. The same EC→iGC excitability variants as in the pattern-separation experiment were used, each driven by an identical EC input.

In each run, a random half of the iGCs (50 of the 100) received a depolarizing current of 0.5 nA for 5 ms, delivered 400 ms into the recording window, once network activity had stabilized. This injected current I_{ext} enters the neuron as part of the total current I of the Izhikevich model (Equation 1):

$$I = I_{ext} + I_{syn} \tag{15}$$

where I_{syn} is the summed synaptic current (Equation 7) and $I_{ext} = 0.5\,\text{nA}$ during the pulse and zero otherwise. For each variant the protocol was repeated over 30 trials of 10 input patterns, and the resulting 300 runs were pooled for analysis.

The response of each population was quantified with a peri-stimulus time histogram (PSTH) of its firing rate, computed in 1.5 ms bins over the interval from 60 ms before to 60 ms after the pulse. In each bin the population firing rate is the pooled spike count divided by the number of cells, the number of runs and the bin width, and is expressed as a z-score relative to a pre-pulse baseline, taken as the binned population rate from the start of the recording window up to 60 ms before the pulse:

$$z(t) = \frac{r(t) - \mu_{base}}{\sigma_{base}} \tag{16}$$

where $r(t)$ is the population rate in the bin at time t relative to the pulse, and μ_{base} and σ_{base} are the mean and standard deviation of the baseline bins.

To relate the activation to sparsity, as McHugh et al. (2022) did *in vivo*, the Gini index (Equation 13) of each population was also compared, with a paired Wilcoxon signed-rank test over all runs, between the 50 ms windows immediately before and after the pulse.

2.7 Oscillation analysis

To characterize the rhythmicity evoked in BCs by the activation, the post-pulse population rate (5 ms to 60 ms after pulse onset, in 1 ms bins) was detrended by subtracting a Gaussian-smoothed envelope ($\sigma = 8$ ms), and the power spectrum of the residual was estimated by Welch's method (Welch 1967). The peak frequency in the 20 Hz to 200 Hz band was taken as the dominant oscillation frequency (reported for BCs in Online Resource 1).

3 Results

Before the effects of neurogenesis were analysed, it was first verified that the network settles into a stable and biologically plausible behavior. Figure 3a shows the spiking activity of the EC, iGC, mGC and pCA3 populations during a representative simulation. Driven by the EC input, the mGCs fire sparsely and in short, synchronized bouts, with only a small fraction of the population active at any moment. The iGCs are more active, while the pCA3 cells fire at a moderate rate, spread more evenly across the population.

This difference between the two populations can be seen in Figure 3b: the iGC firing-rate distribution is shifted towards higher rates than that of the mGCs, with a median firing rate of 3 Hz against 1 Hz for the mGCs. These values closely match those found *in vivo* by McHugh et al. (2022) (2.3 Hz for adult-born against 1.0 Hz for mGCs), although the shape of the distribution differs. For every cell type, the model's firing rates and population sparsity fell within the ranges compiled from the literature (Online Resource 1, Table S1), which is taken as evidence that the network behaves plausibly.

3.1 Adult neurogenesis enhances pattern separation in the DG and CA3

Figure 4a shows that, with neurogenesis, the mGC population activation drops, and with each increasing level of iGC excitability the mGC activation decreases further, from 7.9% in the control network to 3.6% at the highest excitability; the opposite is true for the iGC population, since higher levels of excitability are expected to make the iGCs more active, with almost 90% of the iGC population being active for the highest level of excitability simulated. This happens because of the feedforward inhibition

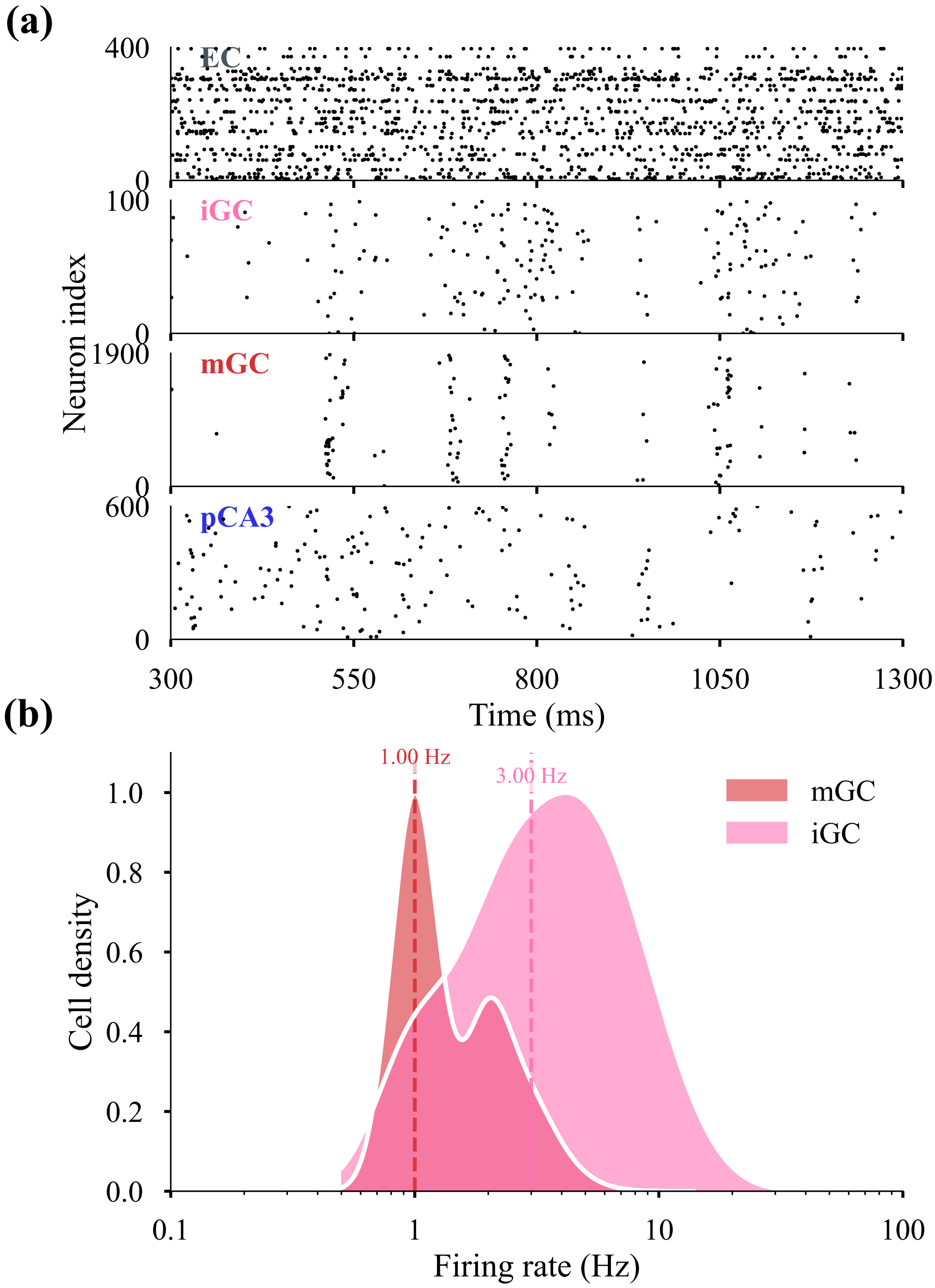

Fig. 3: Baseline network activity. **(a)** Raster plot of the spike times of the EC, iGC, mGC and pCA3 populations in a representative simulation, over the 300 ms to 1300 ms recording window (the first 300 ms discarded for stabilization). **(b)** Kernel-density estimate of the per-cell firing-rate distribution in logarithmic scale of the mGC and iGC populations; the dashed lines mark the population medians (mGC, 1 Hz; iGC, 3 Hz).

exerted by iGCs upon mGCs, via iGC → mGC, iGC → BC/HIPP → mGC and iGC → MC → BC/HIPP → mGC. For increasing levels of iGC excitability, as mGC activity decreases and iGC activity increases, an equilibrium of approximately 8% of the entire GC population (iGCs and mGCs counted together) stays active, similarly to the control model without neurogenesis.

This lower activation seen in the mGC population with adult neurogenesis directly affects pattern separation: with fewer neurons active for a given pattern, the sparser coding leads to an increase in the mGC pattern separation degree, which rose monotonically with iGC excitability from $\mathcal{S}_D = 2.40$ in the control network to 5.80 at full excitability (Spearman $\rho = 1.00$, $p < 10^{-6}$; Figure 4b). Resolved by input similarity, the mGC separation showed the same ordering at every level, being strongest for the most similar inputs and higher than the control across the whole similarity range (Online Resource 1, Figure S1). The two granule-cell types, however, behave in opposite ways. Taken on their own, the iGC outputs never separate their inputs ($\mathcal{S}_D < 1$, falling from 0.45 to 0.20 as excitability rises), acting instead as pattern integrators, as also reported by Kim and Lim (2024b). When the whole granule-cell population is considered together, pattern separation remains above one but decreases with iGC excitability (from $\mathcal{S}_D = 2.40$ to 1.22; $\rho = -0.96$, $p < 10^{-5}$). Because the total granule-cell activation stays near 8% across excitability levels, this decline is driven almost entirely by a loss of output decorrelation, the mean pairwise correlation between output patterns rising from 0.14 to 0.55, rather than by any change in activation. Since the role of iGCs is thought to be the modulation of network activity rather than the encoding of information (McHugh et al. 2022; Fölsz et al. 2023), the relevant quantity is the separation achieved by the encoding population, the mGCs, which is enhanced by neurogenesis.

The same analysis applied to the pCA3 population (Figures 4c and 4d) shows that its mean activation also decreases with neurogenesis and iGC excitability, from 29.7% to 24.1%, while its separation degree increases in the same direction as that of the mGCs, from $\mathcal{S}_D = 0.51$ to 0.69 ($\rho = +0.98$, $p < 10^{-5}$). Unlike the DG, however, the pCA3 separation degree remains below one throughout: by the definition described in Equation 12, the CA3 operates in the convergence regime, its outputs more alike than its inputs. This is expected of a region whose recurrent connectivity supports autoassociation and pattern completion (Rolls 2013; Kopsick et al. 2024) rather than separation. Neurogenesis therefore does not turn the CA3 into a separator, but reduces its convergence.

3.2 Enhanced separation is driven mainly by increased sparsity

Figure 4e shows that the sparsity indices of both the DG and the CA3 increase with neurogenesis and with the progressive excitation of the iGCs, reproducing the central result of McHugh et al. (2022), where adult-born granule cells promote population sparsity throughout the hippocampus. Across the ten excitability levels, the Gini index rose monotonically for the whole granule-cell population (from 0.93 to 0.96; Spearman $\rho = +0.99$, $p < 10^{-6}$) and for the pCA3 cells (from 0.74 to 0.78; $\rho = +0.99$, $p < 10^{-6}$), each above its control value. The Gini index is shown here; the Hoyer measure, computed for cross-validation, follows the same trend (Online Resource 1, Figure S2).

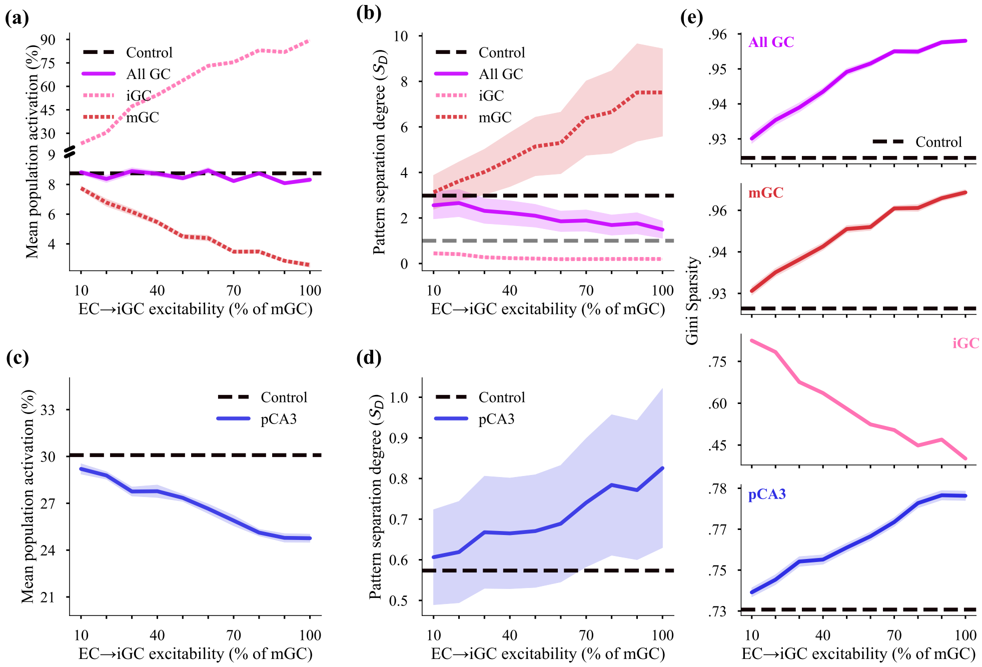

Fig. 4: Effects of adult neurogenesis and iGC excitability on activation, pattern separation and sparsity in the DG and CA3. In every panel the horizontal axis is the EC→iGC excitability level (10–100%), each point is the mean over the 30 pattern sets, shaded bands show the SEM and the dashed black line marks the control network without neurogenesis. **(a)** Mean population activation of DG GCs. **(b)** Pattern separation degree $\mathcal{S}_D$ of DG GCs; the grey dashed line marks $\mathcal{S}_D = 1$, above which the output patterns are more distinct than their inputs. **(c)** Mean population activation of the pCA3 cells. **(d)** Pattern separation degree of the pCA3 cells. **(e)** Population sparsity (Gini index) of the DG GCs and pCA3 cells.

The separation degree depends on both the orthogonalization degree and the activation degree of the patterns (Equations 8–12), so the mGCs could separate better by making their outputs sparser, more decorrelated, or both. Because the two combine multiplicatively, the change in $\log \mathcal{S}_D$ splits into an additive contribution from each. Most of the increase (about 92%) comes from the lower activation degree, as fewer mGCs fire for each pattern. The remaining 8% comes from a smaller but real gain in orthogonalization, as the mGC outputs become less correlated (Online Resource 1, Figure S3); this is not merely an effect of the sparser activity, since it remains when the number of active cells is held fixed. This extra decorrelation is likely due to the stronger winner-take-all competition driven by the iGCs (Coultrip et al. 1992). The entire GC population behaves the opposite: its decline in separation (Section 3.1) stems almost entirely (about 97%) from the increased correlation among its outputs, rather than from any change in activation.

3.3 iGC activation recruits circuit-wide inhibition

Directly injecting current into half of the iGCs propagated through the entire DG–CA3 network, as summarized by the PSTHs of Figure 5, pooled over all EC→iGC excitability variants.

Within the DG, the MCs were strongly excited and in turn drove strong excitation to the BCs. The BCs responded with a large, significant increase in firing, organized as a synchronous population rhythm at $\sim 123\,\mathrm{Hz}$. The fast-spiking BCs are able to sustain a rhythm of this frequency, as their firing rate reaches up to $\sim 230\,\mathrm{Hz}$ *in vivo* (Lübke et al. 1998).

The HIPPs, by contrast, were only weakly inhibited (peak $z \approx -1$), a counter-intuitive outcome given that they receive excitatory input from both the iGCs and the MCs; a plausible cause is that the transient drop in overall GC output following the pulse withdraws the GC→HIPP excitation that would otherwise engage them, outweighing the direct MC drive.

The mGCs displayed a biphasic response to the activation of the iGCs: a small excitation at first (peak $z \approx 4$), followed by an even smaller inhibition afterwards, with the net effect over the 60 ms being excitatory.

In CA3, the iCA3 interneurons were strongly and significantly activated, and this recruitment translated into a small inhibition of the pCA3 cells through the iCA3→pCA3 projection. McHugh et al. (2022) reported this interneuron recruitment to follow iGC spikes with a lag of 20 ms to 40 ms *in vivo*; because axonal conduction delays were not modeled and all synaptic delays were fixed at 1 ms, the absolute latencies here are shorter and not directly comparable, although the temporal ordering of the responses is preserved.

Comparing the Gini index of each population between the 50 ms windows immediately before and after the pulse confirmed that this recruitment of inhibition increased population sparsity in most of the models, the within-trial counterpart of the steady-state sparsification reported in Section 3.2; the full comparison across populations and excitability levels is given in Online Resource 1, Figure S4. Overall, direct iGC activation recruited inhibitory interneurons across both the DG and CA3.

4 Discussion

Taken together, and in agreement with McHugh et al. (2022), activating the iGCs recruited interneurons in both the DG and CA3, providing the circuit mechanism for the rise in granule-cell population sparsity reported in Section 3.2. The model departs from their *in vivo* findings in two respects: the sign of the mGC response, a net excitation rather than the suppression expected of a principal cell, and the small inhibition of the pCA3 cells, for which McHugh et al. (2022) found no effect. Their area-level recordings, however, could not separate the inhibitory contribution of the HIPPs from that of the BCs, so the divergence between the two DG interneuron populations cannot be adjudicated against their data.

The net excitation of the mGCs is itself informative. That neither the direct iGC→mGC inhibitory synapse nor the feedforward and feedback inhibition triggered by the pulse were able to suppress the mGCs indicates that the model captures

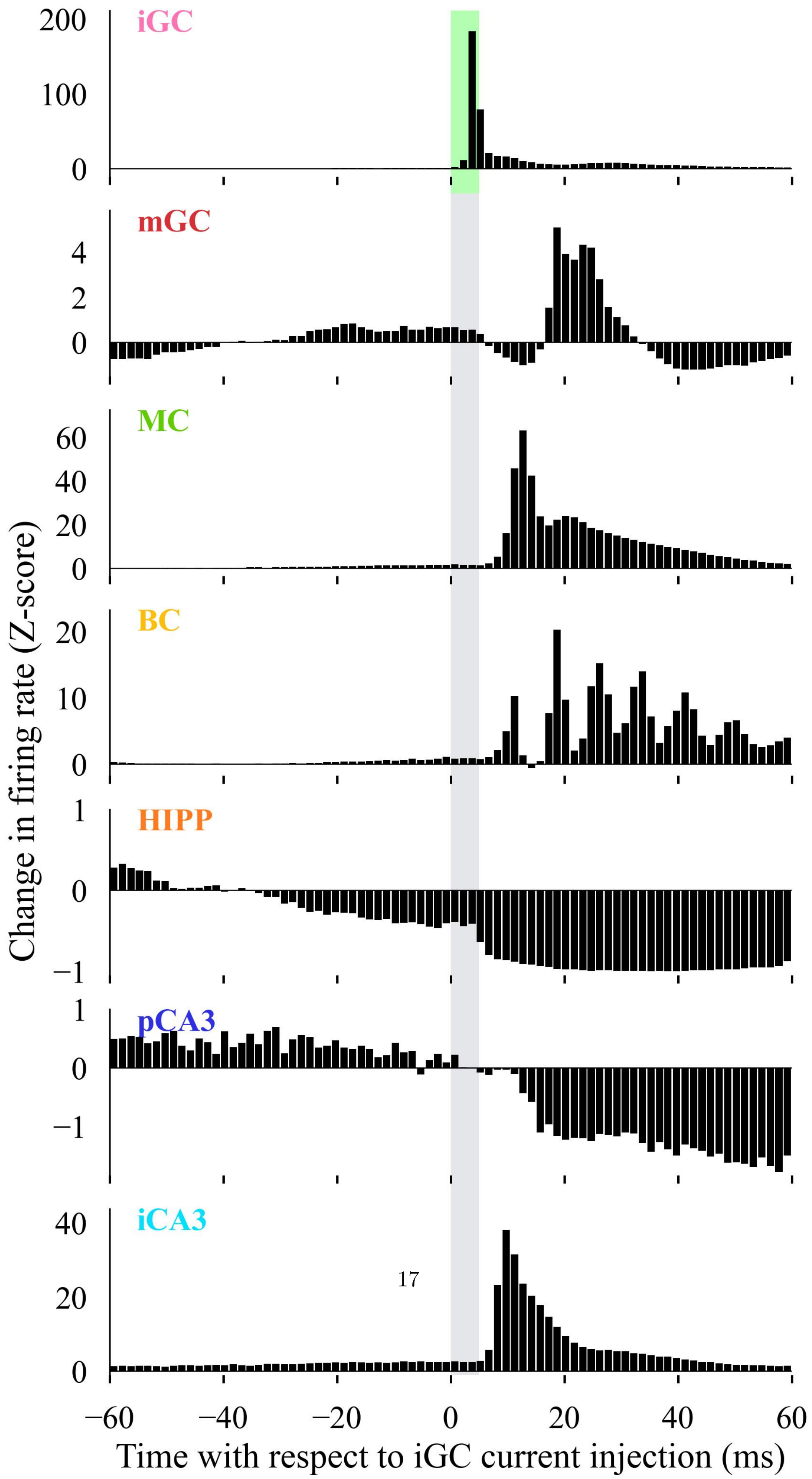

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Fig. 5: Circuit-wide response to direct iGC activation. PSTHs of the population firing rate for each cell type, expressed as a z-score relative to a pre-pulse baseline (Equation 16) and aligned to the onset of the 5 ms depolarizing current injected into half of the iGCs (shaded band). Bins are 1.5 ms wide and span −60 ms to 60 ms relative to the pulse; data are pooled over all EC→iGC excitability variants and all trials. Vertical scales differ across panels. The directly driven iGCs (top) dominate during the pulse; downstream, the MCs, BCs and iCA3 cells show strong excitation, the pCA3 cells and HIPPs a weaker inhibition, and the mGCs a small biphasic response.

the iGC→mGC interaction only imperfectly, and points to a missing inhibitory contribution, plausibly from interneuron types not represented in the reduced circuit.

Supplementary information. TODO

Acknowledgements. Not applicable.

Declarations

Not applicable.

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Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.

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